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Research Article

Keywords: Prevalent chronic disease, Dental Caries, YOLO-V8

Posted Date: December 23rd, 2024

DOI: <https://doi.org/10.21203/rs.3.rs-5657504/v1>

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Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at BMC Oral Health on December 20th, 2025.
See the published version at <https://doi.org/10.1186/s12903-025-07486-x>.

Examination of new clinical dental caries in school children using real intra oral photos with artificial intelligence model YOLO-V8x

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Abstract

Background: Dental caries is one of the most common chronic diseases in school-aged children, with a prevalence above 80% in Indonesia. Traditional diagnostic practices are time-consuming and dependent on the number of healthcare professionals available, and as such, have led to the creation of AI-based alternatives, such as the HI Bogi application. This research used the YOLO-v8 model to assist in the detection of dental caries, which is faster, more efficient, and highly accurate, and can be developed to enhance existing dental health programs in schools in Indonesia

Materials and Methods: A dataset of 3,221 JPG images labeled using the ICDAS D0 to D1 method was prepared and processed using Roboflow labeling software, and the images were resized to 640×640 pixels to standardize the input for model training. The dataset was divided into training (2,266 images), validation (635 images), and testing (320 images) subsets. The YOLOv8x algorithm was used for deep learning and the performance of the model was evaluated using confusion matrix analysis to calculate True Positive (TP), False Positive (FP), and False Negative (FN) values. Statistical Mann–Whitney tests were conducted to compare the classification accuracy between the AI model and dentists across the ICDAS categories, whereas a diagnostic speed test assessed the efficiency of the AI model relative to dentists.

Results: The YOLO-v8 model showed encouraging results, with a recall of 41.1%, accuracy of 72.6%, and mAP of 45.8%. P-values of 0.301 for D1, 0.690 for D2, 0.621 for D3, 0.693 for D4, 0.634 for D5, and 0.302 for D6 were obtained from comparative testing with dentists. These results showed no significant variations in the sensitivity, specificity, PPV, or NPV among the ICDAS categories ($p > 0.05$). Furthermore, AI performed much faster than the dentists during the examinations ($p = 0.000$). AI may improve the efficiency and efficacy of early caries diagnosis as demonstrated by these findings.

Conclusion: The YOLO-v8 model integrated into the application showed promising results in early caries detection, comparable to those of dentists across all ICDAS criteria. AI significantly outperformed dentists in terms of examination speed and completed tasks four times faster. Future research should explore transformer-based models to improve accuracy and expand the datasets to enhance the ability of the model to identify diverse caries categories, including rare lesions.

Keyword :

- Prevalent chronic disease
- Dental Caries
- YOLO-V8

Background

Dental caries, commonly known as tooth decay, is one of the most prevalent chronic diseases in school-aged children worldwide. Based on Indonesian survey, the prevalence of dental caries among children is more than 80%.¹ Despite significant advancements in dental care, early detection and effective intervention remain critical to prevent severe dental complications and improve oral health outcomes. Traditional diagnostic methods often rely on clinical examinations, which, although effective, can be time-consuming and require highly trained professionals.² Therefore, early detection of caries is crucial to reduce treatment costs and time, as well as to prevent further complications. In this context, the development of more detailed and equitable school-based dental health programs across Indonesia is urgently needed to prevent oral diseases, including caries. One of the programs that has long been implemented is the school dental health program (*Usaha Kesehatan Gigi Sekolah* or UKGS). Although it has novel objectives, the implementation of this program has not been optimal, leaving many students without adequate dental health services.

One of the main challenges in screening caries status in elementary schools is the limited time and number of healthcare professionals, such as dentists and dental nurses, at community health centers (*Puskesmas*). To address this issue, the HI Bogi application was developed, enabling independent caries status screening through mobile devices. However, this application still requires results interpretation by dentists, which takes a considerable amount of time, thus posing challenges to its efficiency.^{3,4}

As a solution, the application of Artificial Intelligence (AI) based on the Internet of Dental Things (IoDT) to dental health can be a significant innovation. AI can mimic human thinking processes, allowing caries status assessments to be performed in real time and efficiently. Recent advancements in artificial intelligence (AI) have opened new avenues for enhancing the diagnostic accuracy and efficiency of dentistry. This technology offers the potential for early caries detection, prevention of more severe dental damage, and improvement of the effectiveness of screening in elementary schools. Therefore, research on integrating AI technology into the HI Bogi application needs to be conducted. This study aims to create a system that not only accelerates the caries status assessment process but also ensures that AI assessment results are equivalent to those of dentists, particularly based on the International Caries Detection and Assessment System (ICDAS) classification. Additionally, this research is essential to evaluate the time efficiency of AI-based and conventional methods in caries assessment processes. This integration is expected to be a strategic step in enhancing the effectiveness and efficiency of caries screening programs for elementary school children in Indonesia.³⁻⁷

Among these advancements, object detection models such as You Only Look Once (YOLO) have demonstrated remarkable potential for analyzing medical and dental images. YOLO-v8, as the latest iteration of this model, offers improved accuracy and speed, making it particularly suitable for real-time diagnostic applications.^{8,9}

In this study, we used real intraoral photographs to examine the utility of YOLO-v8 in detecting dental caries among schoolchildren. This approach combines the robustness of AI-

powered diagnostics with practical, noninvasive image acquisition techniques, potentially transforming the traditional caries-detection paradigm. Our investigation aimed to evaluate the performance of this model, highlight its practical applications, and explore its implications for school-based dental health programs.^{8–11}

Material and Methods

Ethical considerations

This study has been approved by the research Ethics Committee of the Faculty of Dentistry, Padjadjaran University with number 1291/UN6.KEP/EC/2023, All procedures involving humans have followed research standards.

Study design

A diagnostic study was conducted using intraoral clinical photographs taken by several research teams to detect caries in patients. The standard for classifying caries in this study uses a classification based on the ICDAS six stages in caries¹². We then trained the deep learning model, YOLO-V8, to classify caries, and evaluated the training results model with dentists using the criteria of sensitivity, specificity, and comparison of dental caries examination time by doctors and artificial intelligence. This study referred to the Checklist for Artificial Intelligence in Medical Imaging (CLAIM)¹³ and STARD standards for reporting diagnostic accuracy¹⁴.

Datasets

In a study collecting the data needed to compile the dataset, researchers found that there are differences in dental characteristics between Indonesian residents and residents of various countries¹⁵. The primary data in this study were collected directly from several locations, including Puskesmas Cimahi Utara, Puskesmas Cimahi Tengah, Puskesmas Cimahi Selatan, SDN Mandiri 02, SDN Mandiri 04, SDN Baros Mandiri Cimahi, SDN Melong Mandiri 04, and SDN Cibabat 2. The research teams were divided into several groups, each consisting of eight dentists, young doctors, and four dental students. The datasets resulted in 3221 intra oral clinical photographs. The next stage is data labeling which is carried out by 3 dentists and one labeling expert, where first one dentist will mark the teeth that are included in the caries tooth classification and enter into d0 - d6, after that, labeling will be carried out by an expert labeler using the labeling application, namely roboflow, after the labeling stage is complete, the results will be validated by two experts who aim to produce a high-quality dataset and ensure that the labeling in the image is appropriate. with the proportion of label distribution on the dental images being similar, as shown in Table 1.

Table 1. Distribution of labels on datasets

Labeled	Instances
D0	1000
D1	900
D2	1200
D3	1100
D4	900
D5	1300
D6	1400

Clasification

The classification of the collected data focused on labeling the data in the dental caries class using guidelines from the Index of Dental Caries and Early Detection (ICDAS) ^{12, 16}. The International Caries Detection and Assessment System (ICDAS) classifies dental caries into seven distinct stages, ranging from D0 to D6 ¹⁷, which reflect the progression of carious lesions from non-cavitated to cavitated forms. Stage D0 indicated a sound tooth surface with no evidence of caries. In contrast, stages D1 and D2 represent early carious lesions, where D1 corresponds to non-cavitated lesions that are visible only under specific conditions, such as drying of the tooth surface, whereas D2 denotes non-cavitated lesions that are visible in a wet environment¹⁸. Stages D3 to D6 signify progressively more severe carious lesions, with D3 indicating a cavitated lesion confined to enamel, D4 representing a lesion that has progressed into dentin but does not yet affect the pulp, and D5 and D6 indicating advanced carious lesions with significant dentin involvement, where D5 is characterized by extensive cavitation and D6 reflects severe carious lesions that may involve the pulp¹⁹. This classification system not only aids in the diagnosis and management of caries but also facilitates communication among dental professionals regarding the severity and treatment needs of carious lesions²⁰. The ICDAS framework has been widely adopted in clinical practice and research, providing a standardized approach to caries assessment that enhances the accuracy of diagnosis and treatment planning²¹. The following are samples of intraoral clinical photographs labeled by the researchers according to the ICDAS classification in Table 2.

Table 2. class label classification guide on datasets with ICDAS method

No	Data images	Class / Caries Stage	Caries Extension
1		D0	No clinically detectable lesion. Dental hard tissue appears normal in color, translucency, and gloss.
2		D1	Earliest clinically detectable lesion compatible with mild demineralization. Lesion limited to enamel or to shallow demineralization of cementum/dentin. Mildest forms are detectable only after drying. When established and active, lesions may be white or brown and enamel has lost its normal gloss.
3		D2	Visible signs of enamel breakdown or signs the dentin is moderately demineralized.
4		D3	Enamel is fully cavitated and dentin is exposed. Dentin lesion is deeply/severely demineralized.
5		D4	No clinically detectable lesion. Dental hard tissue appears normal in color, translucency, and gloss.
6		D5	Earliest clinically detectable lesion compatible with mild demineralization. Lesion limited to enamel or to shallow demineralization of cementum/dentin. Mildest forms are detectable only after drying. When established and active, lesions may be white or brown and enamel has lost its normal gloss.
7		D6	Visible signs of enamel breakdown or signs the dentin is moderately demineralized.

The model

The methodology for utilizing YOLOv8 in the detection of dental caries from imaging data is grounded in its advanced architecture and training techniques, which collectively enhance its performance in real-time applications²². YOLOv8 was built on the CSPDarknet53 backbone, which is an evolution of the traditional Darknet architecture that integrates cross-stage partial (CSP) networks. This design significantly augments the model's learning capacity and efficiency, allowing for improved feature extraction from dental images²³. The architecture comprises three primary components: backbone, neck, and head. The CSPDarknet53 backbone is responsible for extracting hierarchical features, thereby creating a comprehensive visual representation of the input image^{24,25}. The path-aggregation network (PANet) serves as a neck that facilitates effective feature fusion across multiple scales, which is crucial for accurately detecting objects of varying sizes, including early carious lesions²⁶. The YOLO head is tasked with generating bounding box predictions, objectness scores, and class probabilities, which are essential for identifying the presence of caries in dental images²⁷. Furthermore, YOLOv8 employs advanced training methodologies, such as Rectified Adam (RAdam) optimization and data augmentation techniques like MixUp, which contribute to faster convergence and enhanced generalization during the training process²⁸. These improvements render YOLOv8 highly adaptable to diverse datasets and application contexts, ensuring its efficacy in various computer vision tasks, including real-time object detection and segmentation, which are particularly relevant for the early detection of dental caries²⁹.

Training and test of the model AI

In the training phase of this study, the Google Colab Pro platform equipped with an A100 GPU and 54GB RAM was used. This platform was chosen because of its superior ability to handle computationally intensive deep learning tasks. In training the deep learning model in this study using a deep learning algorithm or model, namely YOLO - V8x with training using datasets that have been built by researchers show data with ground truth referring to appropriate references and have been validated by experts in the field of dentistry, namely dentists. The Ground truth represents the targeted result in the model learning process and is used to support in the model learning process and is used to support model training, evaluate its performance and report results ³⁰. In this study, the Non-maximum Suppression (NMS) method is used for cases where more than one segmentation is performed. In cases with multiple detection results such as bounding boxes, it was used to filter these results and select the best result ³¹. In this algorithm there are parameters set to produce accurate detection objects with 50% overlap confidence that will eliminate objects with probability lower than 50% so that the detected results are close to the ground truth ³².

The model was trained using a dataset of 640×640 pixel images, comprising 3,221 images divided into three groups: 2,266 images for training, 320 images for testing, and 635 images for validation. This division is crucial for ensuring that the model can generalize well to unseen data, as highlighted in recent literature emphasizing the importance of proper dataset partitioning in machine learning tasks ³³. The training process lasted for 300 epochs, which was expected to provide the model with sufficient time to learn the underlying patterns in the data comprehensively ³⁴.

The optimizer employed during this training was AdamW, which is a variant of the Adam optimization algorithm that incorporates weight decay to mitigate overfitting ³⁵. The learning rate was set to 0.000909, and the momentum was set to 0.9. These hyperparameters are critical for the convergence and stability of the training process, as discussed in various studies that have explored the impact of learning rates and momentum on model performance ^{36,37}. The parameter optimizer was categorized into three distinct groups to address the varying regularization needs of different parameter types. The first group contained 97 weight parameters without weight decay (decay = 0.0), which are typically reserved for parameters that do not require additional regulation ³⁸. The second group included 104 weight parameters with a weight decay of 0.0005, serving as a regulatory mechanism to reduce model complexity and prevent overfitting, which is a common challenge in deep learning ³⁹. The third group comprised 103 bias parameters that were not subjected to weight decay (decay = 0.0), consistent with practices aimed at optimizing the model training ⁴⁰. This parameter division is performed to ensure that each parameter group receives the appropriate treatment so that the training process can run optimally, and the resulting model has good generalization to data that has never been seen before. After the training, the best model is saved. The performance of YOLOv8x is shown in the correlation in Figure 1. Correlogram showing YOLOv8x performance. The size of the dots in the graph represents the confidence value of YOLOv8x for each object. The larger the dot, the higher is the confidence score.

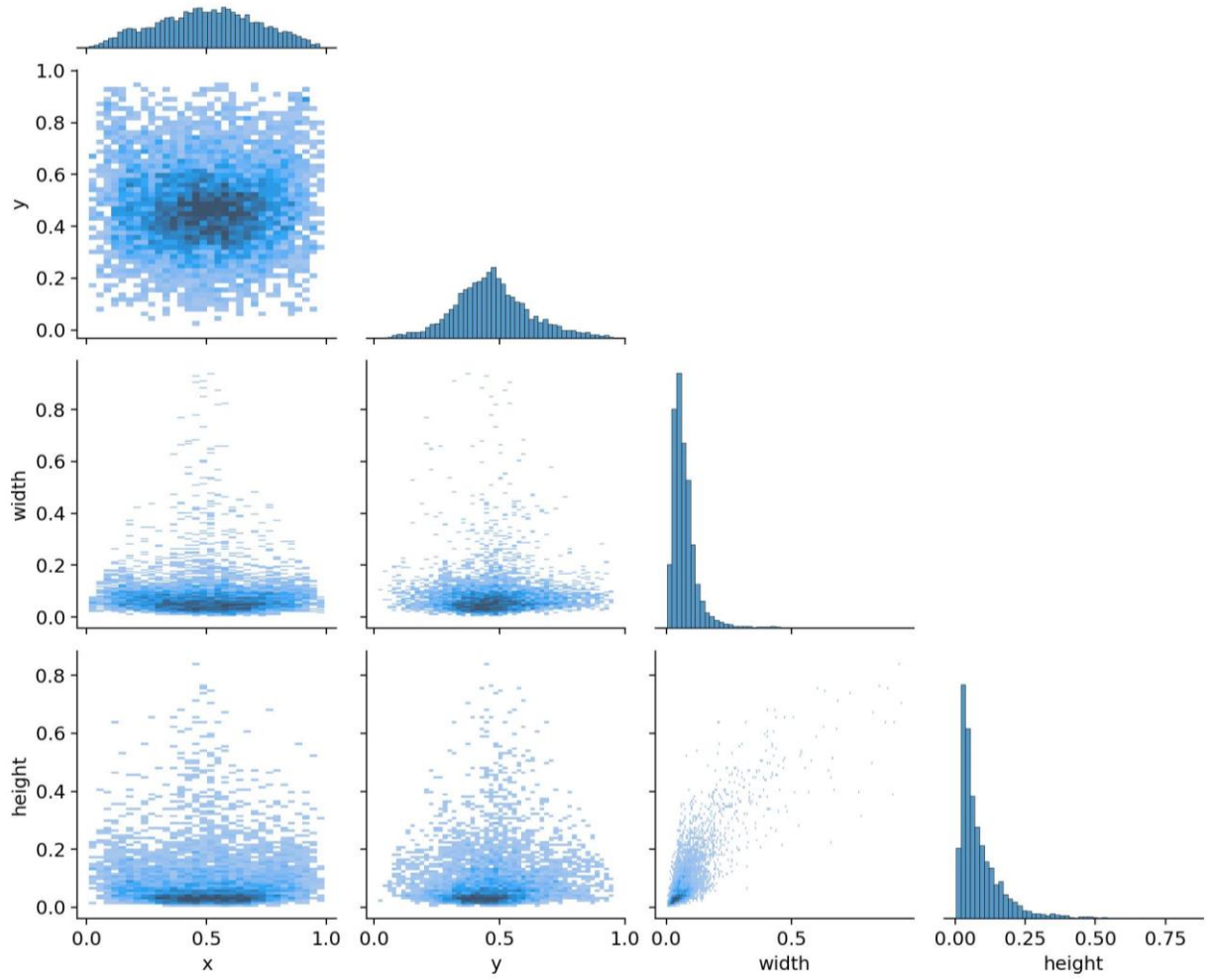


Figure 1. Correlogram showing YOLOv8x performance

Statistical analysis

For model evaluation, several metrics essential for assessing object detection performance were used. These metrics include Intersection over Union (IoU), mean Average Precision (mAP), and Average Precision (AP) ²². The IoU measures the extent to which the bounding box prediction of the model overlaps with the ground truth using the mathematical equation in Equation (1) ⁴¹. mAP, which is the average AP for different object classes, provides an overall picture of the detection accuracy of the model on the dataset used with the mathematical equation in Equation (2) ⁴². The AP itself is a metric that evaluates the precision of the model for each class at various IoU threshold levels, providing a more detailed view of the model's performance in detecting each object type using the mathematical equation in Equation (3).

$$IoU = \frac{(groundTruth \cap prediction)}{(groundTruth \cup prediction)} \quad (1)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i \quad (2)$$

$$AP = \sum_{k=0}^K (R_r(k) - R_r(k + 1)) Rr_{interp}(R_r(k)) \quad (3)$$

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

$$Recall = \frac{TP}{TP+FN} \quad (5)$$

Where:

TP: True positive when the model correctly predicts a positive sample.

FP: false positive; when the model predicts a positive sample, it is negative.

FN: False negative; when the model predicts a negative sample, it is positive.

For comparative evaluation of accuracy between dentists and AI models bivariate analysis used to determine the difference in diagnostic tests for dental caries examination with the Artificial Intelligence system and direct examination by dentists using the Mann-Whitney test ⁴³. Analysis of bivariate data to determine the difference in dental caries examination time with the Artificial Intelligence system and direct examination by dentists using the Mann-Whitney test. The analysis used to determine the difference in detection of dental caries with Artificial Intelligence and direct examination by dentists in elementary school children is the Mann Whitney test ⁴⁴.

Results

The results of this study show that the model trained using the dataset obtained good results, as shown in figure 2. The results of testing using new data show good results, where the labelled and predicted results are the same, with an average confidence above 80%, as shown in Figure 3. The model achieved an MaP value with a 50% IoU of 45.8%, a precision of 72.6%, and a recall of 41.1% (see Table 3). Next, the model was tested again by performing a comparison with dentists using 320 test data points with the label distribution listed in Table 4. The test results showed p-values for D1 = 0.301, D2 = 0.690, D3 = 0.621, D4 = 0.693, D5 = 0.634, and D6 = 0.302 (Table 5), indicating no significant difference (all p > 0.05). This indicates that there was no statistically significant difference between the performance of dentists and artificial intelligence in terms of sensitivity, specificity, PPV, and NPV in each ICDAS category tested. In addition, the model was also tested in terms of examination speed with a p-value of 0.000 (see Table 6), indicating that artificial intelligence (AI) was significantly faster at completing the task than dentists. This indicates that AI can assist doctors in detecting caries more effectively and efficiently.

Discussion

Assessing caries in teeth, especially in the early stages of caries, such as white spot lesions, is quite challenging. There are several ways to assess caries in teeth, one of which is with intra oral photographs ⁴⁵ or with radiographs. Utilizing artificial intelligence technology is quite helpful in assessing early stage caries in teeth with deep learning ⁴⁶. In previous studies, satisfactory results have been found in detecting caries with intelligent systems or AI with the YOLO v3 and

Faster R-CNN models ⁴⁷. In addition, in detecting caries in realtime, good results were obtained using the YOLO v5x model with the results of performance metrics including precision, recall, and F1 score values for detecting white spot lesions are 0.786, 0.618, and 0.692 ⁴⁸. In subsequent research using the YOLO v5 model showed even better results where the highest precision (0.853) occurred in the YOLOv5l model ⁴⁹.

In this study, we found a gap supported by previous research related to the development of AI models for caries stage assessment. Previous research has not used a caries assessment approach with a classification based on the International Caries Detection and Assessment System (ICDAS), which divides caries into seven stages ranging from no caries (D0) to advanced stage (D6). The results of training a dataset with seven caries classes using the YOLO v8x deep learning model showed good results, as shown in figure 2. This model was able to classify caries stages well, especially in the D5 and D6 classes. Late-stage caries are easier for doctors to classify than early stage caries are. In addition, the number of label instances in Table 1 shows that there are more data in the D5 and D6 classes, so the model learns more often in these classes, which contributes to the increased accuracy in assessing advanced caries. Overall, the performance of the model in classifying caries was measured using several metrics, namely, mean Average Precision (mAP), Intersection over Union (IoU), Precision, and Recall, which resulted in values of 45.8%, 50%, 72.6%, and 41.1%, respectively, as shown in Table 3.

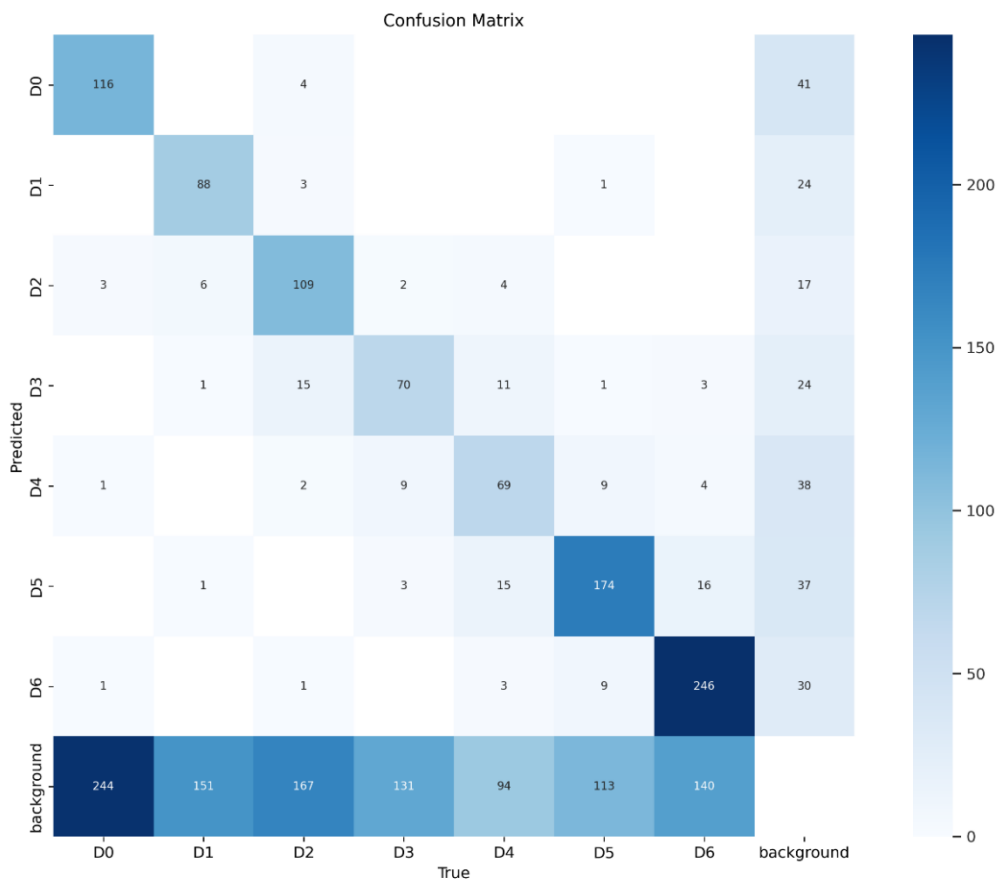


figure 2. confusion matrix

Table 3. YOLO-V8x model training results report with testing using test datasets

ICDAS Classificati on	Average Precision	Mean average precision (mAP)	Intersection Over Union (IOU)	Precision	Recall
D0	39%	45,8%	0,50	72,6%	41,1%
D1	38%				
D2	41%				
D3	33%				
D4	39%				
D5	61%				
D6	66%				

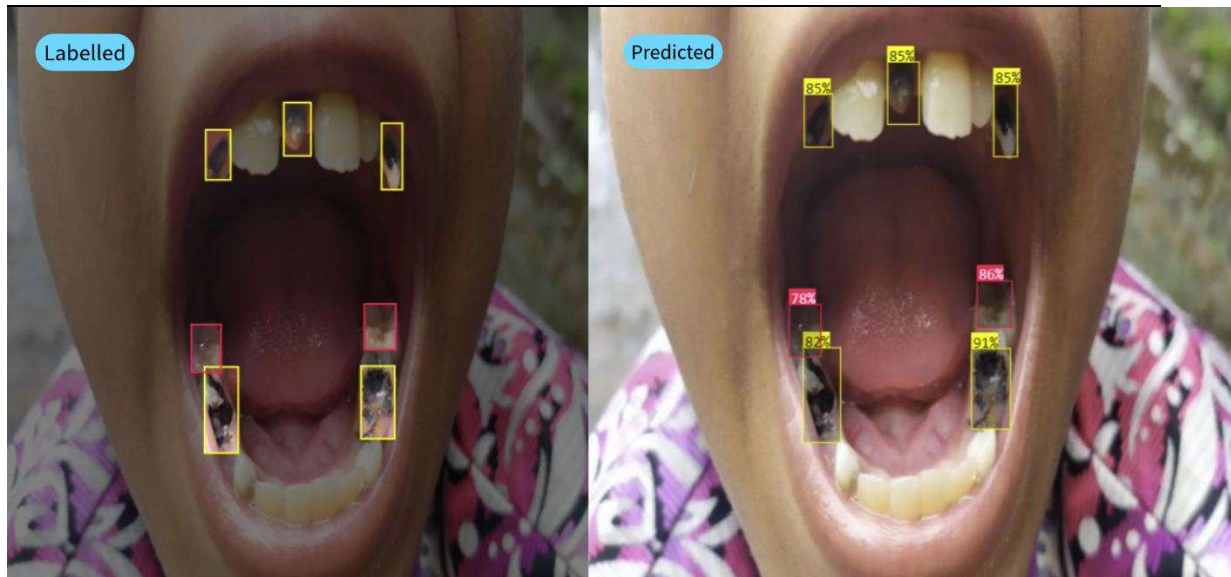


Figure 3. Analysis result on image with bounding box

In figure 4a (precision–confidence curve), the increase in precision as confidence increases indicates the model's ability to reduce false positives at higher confidence levels. A steep curve at low confidence reflects less accurate predictions; however, as confidence increases, the model becomes more confident and produces better precision. The stability of the performance at high confidence (close to 1) indicates that the model is effective in detecting clear data. However, the variation in the lower curve shows that some classes have varying precisions, indicating the model's ability to detect certain classes better than others. In figure 4b (F1-Confidence Curve), the F1 score remains stable in the middle, but decreases at high confidence, indicating a trade-off between precision and recall. This is reinforced by figure 4c (Recall-Confidence Curve), where recall decreases as confidence increases, indicating that the model becomes more selective and misses some positive cases. figure 4d (Precision-Recall Curve) emphasizes this trade-off, where an increase in recall is accompanied by a sharp decrease in precision. Although the model is effective in avoiding errors, the balance between correct positive

detection and correct prediction results needs to be improved, especially in more complex or ambiguous classes.

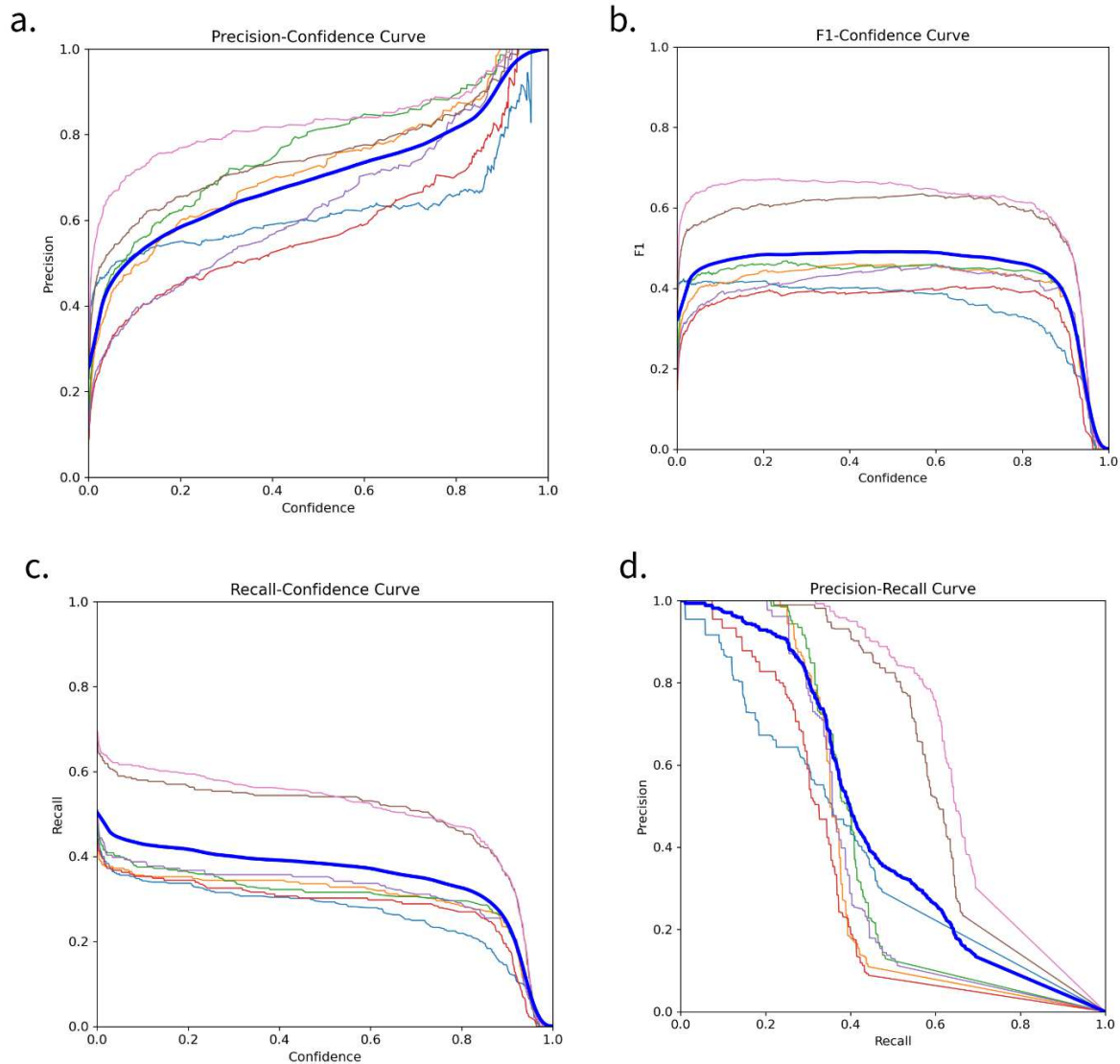


figure 4. Performance graph of training results on YOLO-V8x

After training the YOLO v8x model, a comparative test was conducted to assess the extent to which the results of the AI assessment were in line with those of dentists. This test used 320 intraoral photographs, in which the label frequency of each class varied. Of the 1,486 labels, class D1 had 201 labels (13.52%), whereas class D6 had the highest frequency with 277 labels (18.64%). Other classes, such as D2–D5, had a fairly even distribution, as shown in Table 4.

Table 4. Test data with the frequency of each caries stage label on Cimahi

ICDAS Classification	Frequency	Percentage (%)
D1	201	13,52
D2	270	18,16
D3	222	14,93
D4	249	16,75
D5	267	17,96
D6	277	18,64
Amount	1486	100

After conducting a comparative test between AI and dentist assessment, the results showed that there was no significant difference between the two methods in detecting and classifying caries at various stages of the ICDAS. Based on the data in Table 5, the P-value for each ICDAS class was greater than 0.05, indicating that the difference in results between AI and dentist assessment was not statistically significant. In line with the study, Moharrami et al. noted that various image processing techniques were used in the AI model, yet performance for different lesion severities remained consistent across studies, suggesting that AI can match dentists' judgment in certain contexts⁵⁰. This is consistent with findings from Devlin et al. who reported that AI systems served as an effective adjunct to traditional diagnostic methods, but the difference in diagnostic results was not statistically significant when compared to experienced dentists⁵¹.

A comparison of the performance of AI and dentists in detecting caries in each ICDAS class from D1 to D6 shows that the difference between the two methods is not significant in line with the findings in the research of Shaya et al⁵², and Similarly, Güneç's research demonstrated that AI significantly increased the accuracy of caries detection, particularly in bitewing radiographs, but did not establish a statistically significant superiority over dentists' evaluations⁵³. In the early caries classes (D1 and D2), the sensitivity and specificity of AI were close to those of dentists, with P values greater than 0.05, indicating that AI was able to detect early stage caries with almost equal accuracy to dentists. In the D1 class, the sensitivity of AI (90.17%) was slightly higher than that of dentists (89.80%), whereas in the D2 class, AI had a slightly lower sensitivity (82.31%) than doctors (82.84%), but this difference was not statistically significant.

AI also performed competitively with high sensitivity and specificity in more advanced caries classes (D3–D6). At the D3 grade, the sensitivity of AI (83.12%) was very similar to that of dentists (83.53%). In classes D5 and D6, AI showed excellent performance, with a positive predictive value (PPV) almost identical to that of doctors, especially in class D5 (99.55% for AI and 99.43% for doctors). These results suggest that AI is superior in detecting late-stage caries, particularly on D5 and D6, which is supported by the availability of more data in these classes. Overall, AI provided comparable results to dentists in all ICDAS classes.

The results of the comparison test between the AI-based assessment and dentist's assessment showed that there was no significant difference in detecting and classifying caries at various stages of ICDAS, with all P values greater than 0.05. Previous studies no difference ($p > 0.10$), was noted in caries assessment between telehealth evaluation and clinical examination when conducted by either the dentist or the dental hygienist^{54,55}. AI models, particularly YOLO

v8x, were able to achieve sensitivity, specificity, PPV, and NPV levels almost equivalent to those of dentists at each ICDAS grade. This suggests that AI can be a reliable tool for supporting dentists in performing accurate caries diagnosis.

Table 5. Data analysis of the comparative results between the AI Model and dentists

ICDAS	Dentist				Artificial Intelligence				P value
	SEN	SPE	PPV	NPV	SEN	SPE	PPV	NPV	
D1	89,80	90,56	80,83	95,29	90,80	90,17	82,57	95,04	0,301
D2	82,84	84,11	80,45	86,12	82,31	84,47	79,88	86,41	0,690
D3	83,53	71,33	85,59	73,2	83,12	77,10	85,23	74,21	0,621
D4	86,41	84,29	84,12	86,55	86,77	83,87	84,53	86,18	0,693
D5	96,17	99,31	99,43	95,36	96,45	99,33	99,45	95,45	0,634
D6	95,20	90,00	87,42	96,27	95,59	89,36	88,37	96,28	0,302

When the comparison test was performed on the AI model, the dentist in this study also conducted a caries assessment time comparison test with viable time, and the results indicated that there was a significant comparison of caries assessment time, as shown in Table 6. The average time generated by AI was 31.95 seconds with a standard deviation (SD) of 3.2 seconds, while the time taken by dentists to perform caries assessment was 133.61 seconds with a standard deviation of 6.53 seconds. This difference has a p-value of 0.000, indicating that the difference is statistically significant. These findings are consistent with those of previous studies that highlighted the efficiency of AI in diagnostic processes. For instance, Eschert et al. emphasized that AI can significantly reduce the time required for diagnostic assessments, allowing clinicians to allocate more time to patient care, which is crucial for enhancing patient satisfaction ⁵⁶.

Based on these results, it can be concluded that AI can assist dentists in detecting caries faster than manual assessment by dentists. The speed generated by AI can provide benefits in terms of time efficiency, as well as the potential for increased accuracy in detecting caries at an early stage. This assertion is supported by Schropp's research, which demonstrated that AI-assisted assessments in dental education led to quicker evaluations without compromising the diagnostic accuracy ⁵⁷. Furthermore, Güneç's study indicated that AI tools could enhance decision-making under time constraints, thereby improving diagnostic performance in clinical settings ⁵⁸. The integration of AI into dental practice not only streamlines the assessment process but also holds promise for improving overall diagnostic outcomes, as noted by Khanagar et al., who reviewed the application of AI in the detection and diagnosis of dental caries ⁵⁹.

Table 6. Data analysis of examination time comparison results between AI Models and dentists

	Mean (seconds) ± SD	P value	Interpretation
Artificial Intelligence	31,95 ± 3,2	0,000	Significant
Dentist	133,61 ± 6,53		

Our research shows that the YOLO v8x deep learning model is capable of classifying caries effectively, as evidenced by various testing metrics, including sensitivity, specificity, and a comparative analysis of dental caries examination times between doctors and artificial intelligence. These results indicate that AI can significantly reduce the time required for caries assessment, which is a crucial factor in clinical practice. Future research could focus on doubling the datasets from all regions in Indonesia to create a more varied dataset for detecting caries stages. This approach aligns with findings from Alshaya et al., who noted that variations in data can influence the effectiveness of AI in clinical applications ⁶⁰. Moreover, expanding the dataset is essential for improving the robustness of AI models, as highlighted by Zhu et al., who found that a larger and more diverse dataset leads to improved diagnostic accuracy in dental radiographs ⁶¹.

Conclusion

The study demonstrated that the application of the YOLO-v8 AI model achieved promising results for early dental caries detection based on ICDAS classifications, with MaP value with 50% IoU of 45.8%, 72.6%, and 41.1 %, respectively. Comparative analysis revealed no statistically significant differences between the model and dentists in the detection of caries across all ICDAS criteria (D1–D6). However, AI significantly outperformed dentists in terms of examination speed and completed tasks four times faster. Application of the model to elementary school children in Cimahi identified D2 lesions, representing early caries, as the predominant category, highlighting a pressing public health concern. For future research, development could focus on using transformer-based AI models, which are known to be better at understanding complex patterns, to improve caries-detection accuracy. In addition, increasing the number of datasets with a wider variety will allow the model to learn more about caries categories, including lesions with unique characteristics that are rare.

Acknowledgements

The authors wish to thank DRPM Universitas Padjadjaran for supporting this study. This paper and the research behind it would not have been possible without the exceptional support of IT programmer Rifaz Muhammad Sukma. His knowledge and skills in developing this Artificial Intelligence model were excellent.

Author contributions

Rina Putri Noer Fadilah: study design; collection of data; data analysis/interpretation; writing of the manuscript. Arlette Suzy Setiawan: study design; writing of the manuscript. Rasmi Rikmasari: study design; writing of the manuscript. Saiful Akbar: study design; writing of the manuscript.

Funding

None.

Data availability

Data are available from the corresponding authors.

Declarations

Conflict of Interest

The authors declare that they have no conflicts of interest.

Consent for publication

Not applicable.

Informed consent

Informed consent was obtained from all the participants.

Ethics approval and consent to participate

This study has been approved by the research Ethics Committee of the Faculty of Dentistry, Padjadjaran University with number 1291/UN6.KEP/EC/2023, All procedures involving humans have followed research standards.

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